

05. Pelvic Organ Prolapse: Symptoms, Compartments, POP-Q, and Nonsurgical Care

Companion reading handout for simultaneous listening.

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Pelvic organ prolapse teaches one of the most important principles in office gynecology:

Anatomy alone is not disease.

You can examine two patients and see similar descent. One patient says, I had no idea, nothing bothers me. The other says, I feel a bulge every day, I avoid walking, sex is painful, I have to push the bulge back to urinate, and bowel movements are exhausting.

Those are not the same clinical problem.

The anatomy may look similar. The disease burden is completely different.

So the dominant model for this lecture is:

Treat the symptom, function, anatomy problem. Not anatomy by itself.

Ask five questions.

What does she feel?

What function is impaired?

Which compartments are involved?

Where is the apex?

And what does she want from treatment?

That model prevents two opposite errors. It prevents you from ignoring a patient whose prolapse is deeply affecting her life. And it prevents you from turning an incidental exam finding into unnecessary treatment.

The definition

Pelvic organ prolapse is descent of the anterior vaginal wall, posterior vaginal wall, uterus or cervix, or the vaginal apex after hysterectomy. That descent allows nearby organs to herniate into the vaginal space.

Older words are familiar: cystocele, rectocele, enterocele, uterine prolapse, vault prolapse.

Those words are useful, but they can also mislead. They name the organ you think is behind the vaginal wall. POP-Q and compartment language are more disciplined. They describe what you can actually measure: anterior wall, apex, posterior wall, opening, and length.

Mild descent is common. ACOG is explicit: mild pelvic descent should not automatically be considered pathologic. Pelvic organ prolapse becomes a clinical problem when it causes pressure or bulge symptoms, sexual dysfunction, lower urinary tract dysfunction, or bowel dysfunction.

That is the first resident reflex.

Anatomy describes prolapse. Bother determines treatment.

Now notice the epidemiology, because it explains why this distinction matters.

Only about three percent of women in the United States report vaginal bulge symptoms, but examination-based prevalence can be forty one to fifty percent. That gap tells you something profound. Many anatomic prolapses are asymptomatic.

The peak incidence of symptoms is around age seventy to seventy nine. Women in the United States have about a thirteen percent lifetime risk of surgery for pelvic organ prolapse. And as the population ages, the number of women with prolapse symptoms is expected to rise.

Natural history is also not always dramatic. In one study following symptomatic untreated prolapse for about sixteen months, seventy eight percent of women had no change in the leading edge. In women who do not want treatment, many have no change or only small increase over the next year.

That does not mean prolapse is trivial.

It means the presence of prolapse is not an automatic timer counting down to surgery.

Now, why do symptoms often appear when the prolapse reaches the hymen?

ACOG notes that most women feel symptoms when the leading edge reaches about zero point five centimeters distal to the hymenal ring. That threshold is not random. Above the hymen, descent may still be internal. At the hymen and beyond, tissue enters the outlet space. The patient can feel pressure, rubbing, exposure, chafing, and the mechanical effect of gravity during standing, walking, lifting, and bowel or bladder function.

So when you hear, I feel a bulge, or something is coming out, take it seriously. A vaginal bulge is the symptom most specifically linked to prolapse.

But prolapse is not only bulge.

Lower urinary tract symptoms matter. Ask about stress leakage, urgency, frequency, difficulty initiating voiding, weak or interrupted stream, incomplete emptying, recurrent urinary symptoms, and splinting to void.

Splinting means the patient pushes on or supports the bulging tissue to start or complete urination or defecation. Do not wait for the patient to volunteer this. Many patients are embarrassed, and many do not know the behavior has a clinical name.

Bowel symptoms matter. Ask about straining, constipation, laxative use, fecal incontinence, incomplete rectal emptying, and the need to press inside the vagina or against the perineum to defecate.

Sexual function matters. Ask about dyspareunia, coital incontinence, avoidance of intercourse, and whether the prolapse changes body image or confidence.

These questions are not extra. They determine whether the prolapse is clinically meaningful and which treatment will help.

Pathophysiology

Pelvic support depends on three interacting systems: pelvic floor muscles, fibromuscular connective tissue, and intact innervation.

The levator ani and coccygeus muscles close the pelvic floor and provide lifting forces. The endopelvic connective tissue and vaginal fibromuscular layer attach the vagina and pelvic organs to the pelvic sidewall and apex. The cardinal and uterosacral ligament complex supports the upper vagina and cervix. The distal perineal body and genital hiatus help determine whether the outlet remains closed or widened.

Novak describes this as three levels of support.

Level one is the uterosacral and cardinal ligament complex. Think apex, vaginal length, and vaginal axis.

Level two is the paravaginal attachment of the midvagina to the pelvic sidewall. Think anterior and posterior wall position.

Level three is the distal vagina and perineum. Think outlet, perineal body, and genital hiatus.

This is why the apex matters so much.

The apex is the keystone. When apical support is lost, the anterior wall may descend with it. A patient who looks like she has an anterior wall problem may actually have a major apical support problem. If you only name the bulge as a cystocele, you may miss the support failure driving it.

That is why one of your resident reflexes is:

Always identify the apex.

Risk factors follow from the biology.

Parity and vaginal delivery matter because childbirth can stretch, tear, or denervate pelvic floor muscles and connective support. Novak notes that vaginal delivery roughly doubled the likelihood of prolapse compared with nulliparity in the Women's Health Initiative data. The local Israeli position paper adds forceps delivery and macrosomic delivery as obstetric risk factors.

Age and menopause matter because connective tissue, muscle, estrogen status, and tissue resilience change over time.

Obesity, chronic cough, chronic constipation, and heavy lifting occupations or hobbies matter because repeated intra-abdominal pressure repeatedly loads the pelvic floor.

Connective tissue disorders and genetic predisposition matter because some patients begin with weaker supportive tissue.

So nulliparity is not a risk factor in this framework. It is relatively protective compared with parity.

The evaluation

The most important part of the evaluation is symptom assessment.

Not the stage.

Not the most dramatic looking point.

The symptom assessment.

Record the nature of the bulge, the degree of bother, whether it limits activity or sexual function, whether it is worsening, and what the patient wants from treatment. A patient who wants reassurance is not the same as a patient who wants to walk without pressure, have intercourse without fear, or stop splinting to empty her bladder.

The physical exam should be deliberate.

Start with abdominal and pelvic exam to rule out pelvic masses. Inspect the external genitalia and vaginal epithelium for atrophy, irritation, ulceration, and exposure changes.

Then reproduce the prolapse.

The patient should perform maximal Valsalva. Repetitive coughing can help. If the supine exam does not match the patient's symptoms, repeat the exam standing with the bladder empty. This is a practical point. Gravity is part of the disease experience. A mild supine finding can become obvious when the patient stands.

Use a split speculum examination to isolate compartments.

To evaluate the anterior wall and apex, use one blade to retract the posterior wall. To evaluate the posterior wall and perineal body, turn the blade to retract the anterior wall. The purpose is to avoid looking at a global bulge and pretending you know which compartment is failing.

Assess pelvic floor muscle tone. Can the patient contract and relax voluntarily? Is contraction absent, weak, normal, or strong? Novak gives a practical exam image: palpate the puborectalis and pubococcygeus muscles along the pelvic sidewalls inside the hymen, around the four and eight o'clock positions. A rectovaginal exam may help assess posterior compartment, rectocele, enterocele, anal sphincter tone, and defecatory complaints.

POP-Q

Many residents experience POP-Q as a wall of letters. That is the wrong way to learn it.

Learn it as a map.

The hymen is zero.

Anything above the hymen is negative.

Anything beyond the hymen is positive.

There are four regions: anterior wall, apex, posterior wall, and the opening with length.

On the anterior wall, A a is a fixed point three centimeters proximal to the external urethral meatus. B a is the most descended part of the anterior wall.

So A a is the fixed anterior reference. B a is the moving anterior bulge.

At the apex, C is the leading edge of the cervix or the vaginal cuff.

Point C is crucial. It tells you where the central apical support is. If the uterus is present, C is the cervix. If the patient had a hysterectomy, C is the cuff.

D is the posterior fornix when the cervix is present. It is omitted when the cervix is absent. D helps distinguish cervical elongation from true apical support loss because it reflects the posterior fornix and uterosacral attachment level.

On the posterior wall, A p is a fixed point three centimeters proximal to the posterior hymen. B p is the most descended part of the posterior wall.

Then come the opening and length measurements.

GH is genital hiatus, from the middle of the external urethral meatus to the posterior hymen.

PB is perineal body, from the posterior hymen to the middle of the anal opening.

TVL is total vaginal length, measured with the prolapse reduced.

Here is the memory line:

Anterior wall. Apex. Posterior wall. Opening. Length.

A a and B a.

C and D.

A p and B p.

GH, PB, and TVL.

The reason this becomes clinically powerful is that it makes prolapse shareable. If a patient returns after treatment, or sees another physician, or has recurrent prolapse years later, POP-Q gives a real baseline rather than a vague note saying moderate cystocele.

Stage it

Stages are based on the most distal point relative to the hymen.

Stage zero means no prolapse.

Stage one means the most distal prolapse remains more than one centimeter above the hymen.

Stage two is the hymenal zone: from one centimeter above to one centimeter below the hymen.

Stage three is more than one centimeter beyond the hymen, but not complete or nearly complete eversion.

Stage four is complete procidentia or vault eversion, where the prolapse reaches at least total vaginal length minus two centimeters.

The key memory is that stage two lives around the hymen. That is why stage two often becomes the boundary where anatomy and symptoms start meeting in the patient's lived experience.

Now, do you need additional testing?

Usually, no.

ACOG states that in general, no additional testing beyond complete gynecologic, urologic, and defecatory history and physical examination is needed before treatment.

But there are important exceptions.

If the prolapse extends beyond the hymen, or if the patient has voiding symptoms, record a postvoid residual. Use catheter or ultrasound. Novak notes there is no

universal abnormal cutoff, but if the patient voided at least one hundred fifty milliliters, a residual of one hundred milliliters or less is generally acceptable.

If there is urgency or other lower urinary tract symptoms, minimum assessment includes urinalysis, with culture and microscopy when indicated.

Urodynamics are not routine for every prolapse patient. They may help counseling when there is bothersome incontinence with stage two or greater prolapse, or voiding dysfunction, or when symptoms and exam do not match.

Mechanistically, advanced prolapse can kink or obstruct the urethra. That can cause voiding dysfunction. It can also hide stress urinary incontinence. When the prolapse is reduced by a pessary, speculum, sponge holder, or fingers, the urethra may behave differently. This is the reason cough stress testing with reduction becomes important before surgery, which we will revisit in the surgical lecture.

For now, remember:

If prolapse is beyond the hymen or voiding symptoms exist, measure postvoid residual.

Nonsurgical care

Asymptomatic prolapse

Education and reassurance are appropriate. That is not neglect. It is correct care for a non-bothersome anatomic finding.

But education should be real. Explain what symptoms to watch for: bulge, pressure, worsening voiding, recurrent urinary retention symptoms, splinting, bowel obstruction-type symptoms, ulceration, bleeding from exposed tissue, or quality-of-life limitation.

Second, lifestyle and conservative measures.

Treat constipation. Chronic straining repeatedly loads the posterior compartment and pelvic floor. Fiber and osmotic laxatives may help defecatory dysfunction. Foot elevation while sitting may decrease bulge symptoms in some patients by changing pelvic mechanics and reducing outlet strain.

Address obesity when relevant, not as blame, but as load reduction. Weight does not explain every prolapse, but it is a modifiable pressure factor.

Pelvic floor muscle training can improve symptoms or slow progression, especially in mild to moderate prolapse. It is not a magic anatomic reconstruction. If the prolapse extends well beyond the introitus, response is often lower. But for motivated patients, especially with early symptoms, it can be a meaningful low-risk intervention.

Biofeedback and behavioral retraining may help selected patients with defecatory dysfunction associated with rectocele, especially when the problem is not only anatomy but coordination, straining, and evacuation mechanics.

Local estrogen has limited evidence for treating or preventing prolapse itself. Do not present it as a prolapse cure. But it may help vaginal irritation, atrophy, and pessary tolerance in hypoestrogenic patients.

Pessaries

A pessary is not a consolation prize for patients who cannot have surgery. It is an effective nonsurgical treatment that should be offered to women considering treatment for prolapse.

It is especially attractive for patients who want to avoid surgery, patients with high surgical risk, patients who desire future pregnancy, and patients who want symptom control while delaying definitive repair.

ACOG reports that up to ninety two percent of women can be fitted successfully with a pessary.

Mechanically, a pessary supports the pelvic organs within the vaginal vault. It does not rebuild connective tissue. That is why success depends on fit, retention, comfort, epithelial health, voiding, and the patient's ability or willingness to manage it.

There are support pessaries and space-filling pessaries.

Ring pessaries are common support pessaries. They are easier to fold, insert, and remove. They often work well for stage two and many stage three prolapses.

Gellhorn pessaries are space-filling. They are often used for more advanced prolapse, especially stage four, but they can be harder for the patient to remove and replace.

Novak adds that cube pessaries rely on suction and should generally be removed daily. Donut pessaries are also space-filling devices used for larger prolapse. You do not need every pessary type at your fingertips on day 1, but you do need the core logic: support versus space filling, patient dexterity, vaginal length, introitus, sexual activity, and follow-up capacity.

Fitting is not just inserting a device.

Before fitting, the patient empties her bladder. Estimate size by digital exam and reduction of the prolapse. Choose the type based on anatomy, activity, symptoms, and patient preference. After insertion, the patient should stand, perform Valsalva, and cough. The pessary should stay in place. It should feel comfortable. You should be able to sweep a finger between the pessary and the vaginal wall. And the patient should be able to void with the pessary in place before she leaves.

That last point matters.

If the device corrects the bulge but creates retention or discomfort, you have not fitted it successfully.

Teach self-care when possible. A patient who can remove, clean, and replace the pessary has more independence and can usually have less frequent follow-up.

ACOG recommends annual follow-up for patients who can maintain pessary hygiene themselves. If the patient cannot remove and replace the pessary, regular follow-up every three to four months is needed.

Novak gives a more intensive early fitting pattern: return after one to two weeks, then four to six weeks depending on independence and ability, then extend follow-up based on self-care. This is practical. Early follow-up catches poor fit, discomfort, expulsion, voiding problems, and epithelial irritation.

Pessary complications are usually manageable, but neglect is dangerous.

Common problems include discharge, odor, irritation, failure to retain the device, discomfort from a device that is too large, and vaginal erosion.

ACOG reports erosion in about two to nine percent. Management usually begins by removing the pessary for two to four weeks and using local estrogen when atrophy is present. Resolution may occur even without estrogen. If problems persist, change the size, change the type, or increase follow-up frequency.

Caregivers for patients with dementia or limited self-care ability must understand the follow-up schedule. Rare complications such as fistula, hydronephrosis, urosepsis, and bowel entrapment are usually the result of neglected devices, not careful pessary care.

Let us walk through the clinical logic

Case one.

A sixty eight year old has stage two anterior descent on routine exam. She has no bulge, no pressure, no urinary symptoms, no bowel symptoms, and no sexual bother.

Do not turn the finding into disease. Explain the finding. Document it. Reassure her. Tell her what symptoms should prompt reassessment. Treatment is not required.

Case two.

A fifty nine year old says she feels a bulge by the end of every workday and has stopped walking for exercise. Supine exam shows only mild descent.

Do not dismiss her because your exam was unimpressive. Repeat the exam standing with an empty bladder and maximal strain. The disease happens in gravity and activity, not only in lithotomy.

Case three.

A seventy four year old with prolapse beyond the hymen reports weak stream, incomplete emptying, and pushing the bulge back to urinate.

Measure postvoid residual. Consider urinalysis if symptoms suggest infection. The prolapse may be obstructing or kinking the urethra. Her urinary function is part of the prolapse problem.

Case four.

A patient has posterior compartment prolapse and says she must press in the vagina to complete bowel movements.

That symptom fits posterior compartment dysfunction. But do not assume surgery will fix all constipation. Ask about stool form, straining, laxative use, pelvic floor coordination, fecal incontinence, and pain. Treat constipation and evacuation mechanics. Anatomy and bowel physiology often coexist.

Case five.

A thirty five year old with symptomatic prolapse wants another pregnancy.

Offer a pessary. It can control symptoms and delay surgery until childbearing is complete. This is exactly the kind of patient where nonsurgical treatment is not second-rate; it is aligned with her life plan.

Case six.

An eighty two year old with medical comorbidities wants relief from bulge but wants to avoid surgery.

Again, offer pessary fitting. If she cannot manage it herself, build the follow-up plan before placing it. The device is only safe if the care system around it is real.

Case seven.

A pessary user returns with irritation and a small vaginal erosion.

Remove the pessary. Let the tissue recover for two to four weeks. Consider local estrogen if atrophic. Reassess fit and follow-up frequency. Do not simply reinsert the same device and hope.

Compress the lecture

Pelvic organ prolapse is common, benign, and often asymptomatic. It becomes clinically important when it causes pressure, bulge, sexual dysfunction, lower urinary tract dysfunction, or bowel dysfunction.

The exam language should be POP-Q, because it turns vague words into shared anatomy.

The map is simple:

Anterior wall, apex, posterior wall, opening, length.

A a and B a.

C and D.

A p and B p.

GH, PB, and TVL.

The hymen is zero. Above it is negative. Beyond it is positive.

Stage two is the hymenal zone.

The apex is the keystone.

And treatment begins with the patient's bother and goals.

Reassure asymptomatic patients. Treat constipation and modifiable pressure factors. Use pelvic floor muscle training where it can help. Use local estrogen for atrophy and irritation, not as a prolapse cure. Offer pessary as a real treatment, especially when surgery is undesired, risky, or poorly timed.

The final resident reflex is this:

Map the compartments. Identify the apex. Ask what function is impaired. Measure postvoid residual when prolapse is beyond the hymen or voiding symptoms exist. And offer a pessary before assuming surgery.

That is how prolapse becomes a precise clinical conversation instead of a frightening anatomical label.